

Databases -- Guided Notes

Unit tp-4.5 | CompTIA Network+ N10-009 Obj. FC0-U71 4.5 | N10-008 FC0-U71 4.5

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A database is an organized collection of _____ designed for efficient retrieval and manipulation. A _____ (DBMS) is software that manages the database -- examples include MySQL, PostgreSQL, and Microsoft SQL Server. Users interact with the DBMS using _____ (Structured Query Language).

Question 2

In a relational database, data is stored in _____ (rows and columns). Each row is called a _____ and each column is called a _____. A _____ key is a column whose value uniquely identifies each row. A _____ key in one table references the primary key of another table.

Question 3

The four basic database operations are described by the acronym _____.: _____ (add new records), _____ (retrieve data), _____ (modify existing data), and _____ (remove records).

Question 4

A SELECT query retrieves data from a database. Write the basic SQL syntax to retrieve all records from a table named students:

_____ * _____ students;

To retrieve only students with grade = 'A', add a _____ clause:

_____ * FROM students _____ grade = 'A';

Question 5

Database normalization reduces _____ (storing the same data in multiple places) by organizing tables so that each piece of data is stored _____ once. First normal form (1NF) requires that each column holds _____ values. Normalization improves _____ consistency and reduces the chance of update anomalies.

Question 6

A NoSQL database stores data in formats other than tables, such as _____ (MongoDB), _____ (Redis), _____ (Apache Cassandra), or _____ (Neo4j). NoSQL databases are often preferred when data is _____ -structured, requires horizontal scaling, or schema changes frequently.

Question 7

SQL Injection is a security attack where an attacker inserts malicious _____ into an input field that is then executed by the database. For example, entering ' OR '1'='1 into a login form might bypass authentication. The primary defense is using _____ statements (also called parameterized queries) which treat user input as _____ rather than executable code.

Question 8 -- Real World Application

REAL WORLD: A student builds a web application where users can search for products by name. The search box directly inserts the user's input into a SQL query like: SELECT * FROM products WHERE name = '[input]'. Explain the SQL injection vulnerability, show an example attack, and describe the fix.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-4-5-databases/tp-4-5-databases-answer-key.pdf
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